Yuzhou Joey Zou

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Research Interests

Microlocal Analysis, Inverse Problems, Partial Differential Equations, Mathematical Physics.

Employment

Dates	*Position*
2025–	Oakland University, Assistant Professor
2022–2025	Northwestern University, Boas Assistant Professor
	Postdoctoral mentor: Jared Wunsch
2021–2022	University of California, Santa Cruz, Postdoctoral Scholar-Employee
	Postdoctoral mentor: François Monard

Education

Dates	*Institution and Degree*
2016–2021	Stanford University, Ph.D. in Mathematics
	Thesis advisor: András Vasy
	Thesis title: *Microlocal analysis with applications to seismic inverse problems*
2012–2016	University of Chicago, B.S. with honors in Mathematics and B.A. in Chemistry
	GPA: 3.83

Publications

[14] "**The asymptotic structure of forward scattering**", joint work with Nicholas Lohr, Izak Oltman, and Ethan Sussman. Preprint, 2026. arXiv:2608.27815

[13] "**Gaussian-weighted normal operators on Euclidean space**". Preprint, 2026. arXiv:2605.02232

[12] "**A Comparison Test for Meromorphic Extensions**", joint work with Adi Glücksam. Preprint, 2026. arXiv:2601.05896

[11] "**Tensor tomography on asymptotically hyperbolic surfaces**", joint work with Nikolas Eptaminitakis and François Monard. Preprint, 2025. arXiv:2510.04144.

[10] "**A Gutzwiller trace formula for singular potentials**", joint work with Jared Wunsch and Mengxuan Yang. Preprint, 2025. arXiv:2509.05220.

[9] "**Asymptotic Expansion of the Eigenvalues of a Bathtub Potential with Quadratic Ends**". *Communications in Mathematical Physics*, Vol. 406, no. 220. 2025. DOI: 10.1007/s00220-025-05394-3

[8] "**The hyperbolic X-ray transform: new range characterizations, mapping properties and functional relations**", joint work with Nikolas Eptaminitakis and François Monard. *Inverse Problems and Imaging*, Vol. 22, 65-100. 2026. DOI:10.3934/ipi.2025045

[7] "**Helmholtz quasi-resonances are unstable under most single-signed perturbations of the wave speed**", joint work with Euan A. Spence and Jared Wunsch. *Journal of Differential Equations*, Vol. 440, Part 2. 2025. DOI:10.1016/j.jde.2025.113441.

[6] "**The Morse index theorem for mechanical systems with reflections**", joint work with Jared Wunsch and Mengxuan Yang. *Nonlinearity*, Vol. 37, no. 8. 2024. DOI: 10.1088/1361-6544/ad5636.

[5] "**Boundary triples for a family of degenerate elliptic operators of Keldysh type**", joint work with François Monard. *Pure and Applied Analysis*, Vol. 6, no. 2, 541-580. 2024. DOI: 10.2140/paa.2024.6.541.

[4] "**The C^∞-isomorphism property for a class of singularly-weighted X-ray transforms**", joint work with Rohit K. Mishra and François Monard. *Inverse Problems*, Vol. 39, no. 2. 2023. DOI: 10.1088/1361-6420/aca8cb.

[3] "**Microlocal Methods for The Elastic Travel Time Tomography Problem for Transversely Isotropic Media**". Preprint, 2021. arXiv:2112.14455.

[2] "**Streak artifacts from non-convex metal objects in X-ray tomography**", joint work with Yiran Wang. *Pure and Applied Analysis*, Vol. 3, no. 2, 295-318. 2021. DOI: 10.2140/paa.2021.3.295.

[1] "**Partial Global Recovery in the Elastic Travel Time Tomography Problem for Transversely Isotropic Media**". *Annales de l'Institut Fourier*, Vol. 74, no. 5, 2077-2139. 2024. DOI: 10.5802/aif.3617.

Expository Papers

[2] "**Entropy and kinetic formulations of conservation laws**".
Written at the University of Chicago Mathematics REU 2015.

[1] "**Modes of convergence for Fourier series**".
Written at the University of Chicago Mathematics REU 2014.

Awards and Honors

Date	*Award*
2026	Faculty Research Fellowship, Oakland University
2021	Mathematics Distinguished Service Award, Dept. of Mathematics, Stanford University
2019	Robert Osserman Teaching Award, Dept. of Mathematics, Stanford University
2018, 2016	Honorable Mention, NSF Graduate Research Fellowship
2016	Paul R. Cohen Memorial Prize, University of Chicago Dept. of Mathematics Awarded to top graduating mathematics majors
2016	1st prize at the 23rd International Mathematics Competition, Blagoevgrad, Bulgaria (19th place overall)
2015	Honorable Mention, Putnam Exam

Teaching

Instructor, Oakland University

Duties: write and give lectures, write homework and exams.

Term	*Year*	*Course*	*Course Title*
Winter	2027	APM 2555	Intro to Differential Equations with Matrix Algebra
Winter	2027	MTH 5551	Real Analysis
Fall	2026	MTH 3002	Intro to Advanced Mathematical Thinking
Summer	2026	MTH 1441	Precalculus
Winter	2026	MTH 1554	Calculus 1
Fall	2025	MTH 1554	Calculus 1

Instructor, Northwestern University

Duties: write and give lectures, write homework and exams.

Term	*Year*	*Course*	*Course Title*
Spring*	2025	Math 228-1	Multivariable Differential Calculus for Engineering
Winter	2025	Math 220-2	Single-Variable Integral Calculus
Spring*	2024	Math 220-2	Single-Variable Integral Calculus
Winter	2024	Math 220-2	Single-Variable Integral Calculus
Winter	2023	Math 230-2	Multivariable Integral Calculus
Fall	2022	Math 220-1	Single-Variable Differential Calculus

* – co-lead instructor

Instructor, University of California, Santa Cruz

Duties: write and give lectures, write homework and exams.

Term	*Year*	*Course*	*Course Title*
Spring	2022	Math 218	Advanced Parabolic and Hyperbolic PDEs
Winter	2022	Math 121A	Differential Geometry

Instructor, Stanford University

Duties: write and give lectures, write homework and exams.

Term	*Year*	*Course*	*Course Title*
Summer$^\diamond$	2021	Math 19	Single Variable Calculus 1
Summer$^\diamond$	2020	Math 19	Single Variable Calculus 1

$^\diamond$ – conducted online

Administrative Teaching Assistant, Stanford University

Duties: manage course logistics for a large ($\sim$ 300 students) course (e.g. arrange exam logistics, manage homework/exam grading, maintain course website, answer student emails, etc.), hold office hours, grade exams.

Term	*Year*	*Course*	*Course Title*
Spring$^\diamond$	2021	Math 51	Linear Algebra and Multivariable Calculus
Winter	2020	Math 51	Linear Algebra and Multivariable Calculus
Autumn	2018	Math 51	Linear Algebra and Multivariable Calculus

$^\diamond$ – conducted online

Teaching Assistant, Stanford University

Duties: lead discussion sections, hold office hours, grade exams.

Term	*Year*	*Course*	*Course Title*
Winter	2018	Math 51	Linear Algebra and Multivariable Calculus

Course Assistant, Stanford University

Duties: hold office hours, grade homework and exams, write solutions.

Term	*Year*	*Course*	*Course Title*
Autumn	2019	Math 205A	Graduate Real Analysis 1
Summer	2019	Math 19	Single Variable Calculus 1
Winter	2019	Math 205B	Graduate Real Analysis 2
Autumn	2017	Math 171	Fundamental Concepts of Analysis
Spring	2017	Math 172	Lebesgue Integration and Fourier Analysis
Autumn	2016	Math 20	Single Variable Calculus 2

Graduate Assistant, Stanford Online High School

Duties: help various aspects of Stanford Online High School operations, including researching high school math curricula, grading for various courses, etc..

Term	*Year*	*Position*
Winter	2021	Research and Teaching Assistant
Autumn	2020	Research and Teaching Assistant

Reader, University of Chicago

Duties: grade homework.

Term	*Year*	*Course*	*Course Title*
Winter	2016	Math 255	Abstract Algebra 2
Autumn	2015	Math 254	Abstract Algebra 1
Spring	2015	Math 205	Analysis in $\mathbb{R}^n$ 3
Winter	2015	Math 204	Analysis in $\mathbb{R}^n$ 2
Autumn	2014	Math 203	Analysis in $\mathbb{R}^n$ 1

Junior Tutor, University of Chicago

Duties: lead discussion section, grade homework.

Term	*Year*	*Course*	*Course Title*
Spring	2014	Math 133	Elementary Functions and Calculus 3
Winter	2014	Math 132	Elementary Functions and Calculus 2
Autumn	2013	Math 131	Elementary Functions and Calculus 1

Conferences, Workshops, and Seminars

Conferences Organized

Date	*Event*
Mar 2026	Special Session on Microlocal Analysis and Mathematical Physics 2026 AMS Spring Eastern Sectional Meeting, Boston College, MA
Jun 2024	Microlocal Analysis and Quantum Dynamics Summer school and conference, Northwestern University

Seminars Organized

Date	*Event*
2023-2024	Analysis Seminar, Northwestern University (co-organized)
Winter 2018	Student Analysis Seminar, Stanford University
Autumn 2017	Kiddie Colloquium, Stanford University

Invited Conference Talks

Date	*Event*
Sep 2026	SIAM Great Lakes Section Meeting, Purdue University
Aug 2026	The VII AMMCS International Conference, Waterloo, Ontario
Jun 2026	Great Lakes Mathematical Physics Meeting, University of Dayton
May 2026	May Midwestern Microlocal Meeting, Chicago
Jan 2026	Special Session on *Recent Trend of Mathematical Modeling and Methods in Medical Applications*, Joint Mathematics Meetings 2026
Jun 2025	Great Lakes Mathematical Physics Meeting, University of Kentucky
Mar 2025	Special Session on *Inverse Problems: Theory and Applications*, 2025 AMS Spring Central Sectional
Dec 2024	Harmonic and Microlocal Analysis in Partial Differential Equations, MATRIX
Oct 2024	Special Session on *Harmonic Analysis, Partial Differential Equations, and Spectral Theory*, 2024 AMS Fall Western Sectional
Oct 2024	Triangle Area Inverse Problems Weekend, NC State University
Aug 2024	Geometric Inverse Problems Summer School, UC Santa Cruz
Jun 2024	Great Lakes Mathematical Physics Meeting, Michigan State University
May 2024	SIAM Conference on Imaging Science, Atlanta
Mar 2024	Ohio River Analysis Meeting, University of Kentucky
Feb 2024	Texas Analysis and Mathematical Physics Symposium, Texas A&M University
Oct 2023	Spectral Theory and Applications, Texas A&M University
Sep 2023	Applied Inverse Problems 2023, Göttingen, Germany
Jun 2023	Special Session on *Inverse Problems and Imaging*, The 13th AIMS Conference on Dynamical Systems, Differential Equations and Applications
Aug 2022	Inverse Problems in Analysis and Geometry, Helsinki
Jul 2022	Workshop on Microlocal Analysis & PDEs, University College London
Dec 2021	Session on *Geometric Tomography and Microlocal Analysis*, 2021 CMS Winter Meeting
Aug 2021	Inverse problems and nonlinearity, Helsinki

Invited Seminar Talks

Date	*Event*
Oct 2026	Colloquium, Oakland University
May 2025	Analysis Seminar, Northwestern University
Mar 2025	Colloquium, Oakland University
Oct 2024	Baby Inverse Problems Seminar (online)
Oct 2024	Inverse Problems Seminar, UC Irvine
Sep 2024	Analysis and Applied Mathematics Seminar, University of Illinois, Chicago
Dec 2023	Spectral and Scattering Theory Seminar, Purdue University
Nov 2023	University College London
Oct 2023	Analysis & PDE Seminar, UC Berkeley
Oct 2023	Geometry & Analysis Seminar, UC Santa Cruz
Oct 2023	Analysis & PDE Seminar, Stanford University
Sep 2023	Analysis & Differential Geometry Seminar, Emory University
Apr 2023	PDE Seminar, Northwestern University
Apr 2023	Inverse Problems Seminar, University of Washington
Oct 2022	Analysis Seminar, Northwestern University
Oct 2022	Geometry and Topology Seminar, NC State University
May 2022	Analysis and PDE Seminar, University of Kentucky
Apr 2022	International Zoom Inverse Problems Seminar, UC Irvine
Oct 2021	Geometry and Analysis Seminar, UC Santa Cruz
Sep 2021	HADES Seminar, UC Berkeley
May 2020	Geometry and Analysis Seminar, UC Santa Cruz
Mar 2020	Differential Geometry & PDE Seminar, University of Washington
Feb 2020	Analysis & PDE Seminar, Stanford University
Feb 2020	HADES Seminar, UC Berkeley
Dec 2019	Graduate Student Seminar, Microlocal Analysis Program, MSRI

Recent Conferences and Workshops Attended

Date	*Event*
Sep 2026	"SIAM Great Lakes Section Meeting", Purdue University
Aug 2026	"The VII AMMCS International Conference", Wilfred Laurier University, Waterloo, Ontario, Canada
Jun 2026	"Great Lakes Mathematical Physics Meeting", University of Dayton
May 2026	"May Midwestern Microlocal Meeting", Chicago
Mar 2026	"AMS Spring Eastern Sectional Meeting", Boston College
Feb 2026	"Modern Applications of Microlocal Analysis", University College London, UK
Jan 2026	"High energy spectral theory: geometry and dynamics", Isaac Newton Institute, UK
Jan 2026	"Joint Mathematics Meeting", Washington, DC
Jun 2025	"Great Lakes Mathematical Physics Meeting", University of Kentucky
Mar 2025	"AMS Spring Central Sectional Meeting", University of Kansas
Dec 2024	"Harmonic and Microlocal Analysis in Partial Differential Equations", MATRIX Institute, Australia
Oct 2024	"AMS Fall Western Sectional Meeting", UC Riverside

Oct 2024	"Triangle Area Inverse Problems Weekend", NC State University
Aug 2024	"Summer School: Geometric Inverse Problems and Inverse Problems for Elliptic Equations", UC Santa Cruz
Jul 2024	"NU Trends in Ergodic Theory", Northwestern University
Jun 2024	"Microlocal Analysis and Quantum Dynamics', Northwestern University
Jun 2024	"Great Lakes Mathematical Physics Meeting", Michigan State University

Service to Profession

Peer reviewer for the following journals:

- Inverse Problems (2 articles refereed)
- Transactions of the American Mathematical Society (1 article refereed)
- SIAM Journal on Mathematical Analysis (1 article refereed)
- Inverse Problems and Imaging (1 article refereed)
- Journal of Functional Analysis (1 article refereed)
- Rocky Mountain Journal of Mathematics (1 article refereed)

Ph.D. defense committees served on:

- Nicholas Lohr, Northwestern University, Apr 2025

Mentoring and Outreach

Dates	*Event*
Summer 2025	Northwestern Dynamics RTG REU, Northwestern University *–Mentored an undergraduate research project on quantum trajectories and quantum-classical correspondences. Served on a panel on "How to Give a Talk."*
Jun 2025	Career Development Panel at GLaMP 2025 *–Served on a panel aimed at early-career mathematicians discussing career development issues and questions at the workshop "Great Lakes Mathematical Physics Meeting 2025."*
Summer 2024	Microlocal Analysis and Quantum Dynamics Summer School *–Organized summer school for undergraduates, graduate students, and early career researchers, covering topics in semiclassical analysis and applications, in preparation for the conference of the same name. Planned logistics for lectures, problem sessions, and social activities for the participants.*
Autumn 2021	Directed Reading Program, University of California, Santa Cruz *–Directed undergraduate reading project in Fourier analysis.*
Autumn 2019 to Spring 2021	TA Mentoring Program, Stanford University (5 quarters) *–Mentored first-time teaching assistants by observing sections and providing feedback.*
Spring 2017 to Spring 2021	Directed Reading Program, Stanford University (9 quarters) *–Directed undergraduate reading projects in Fourier analysis, complex analysis, ergodic theory, geometric measure theory, Ramsey theory, Markov chains, and distribution theory.*
Sept 2020	Workshop on best teaching practices for graduate students, Stanford University

	–Moderated a panel regarding effective strategies for being an effective TA.
Summer 2016	Summer Analysis Bootcamp, University of Chicago
	–Teaching assistant for summer program for advanced undergraduates in analysis.
Summer 2013	Young Scholars Program, University of Chicago
	–Teaching assistant for summer math program for high school students.
Summer 2013	SESAME Program, University of Chicago
	–Teaching assistant for certification program for middle school mathematics teachers.
Spring 2013	Neighborhood Schools Program, University of Chicago
	–Tutor for after-school program at local elementary schools

Other Information

Languages (natural): Mandarin Chinese (native), English (native), Cantonese (basic)
Languages (computer): Python (proficient), LaTeX (proficient)
Citizenship: United States of America